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Algorithmen und Wahrscheinlichkeit
Exercise Session 2

Exercise 1 – *Connectivity*

(a) Assume G is 2-connected. Let (u, v, w) be a path in G of length 2. Show that we can extend this path to a cycle, that is, show that G contains a cycle where u, v, w appear as incident vertices.

(b) Prove that if $G = (V, E)$ is a 2-connected graph, $e \in E$, and $u, v \in V$ is a pair of vertices in G not incident to e, then there is a path in G from u to v passing through e. (*Hint:* in a 2-connected graph, every two edges e, f lie on a common cycle.)

Exercise 2 – *Dirac's Theorem*

In this exercise we will show that if $G = (V, E)$ is a graph with minimum degree $\geq |V|/2$, then G has a Hamiltonian cycle.

(i) We want to show that for any path $v_1, \dots v_k$ in G (with $k < n$), we can extend it. Assume that $v_1, \dots v_k$ is a path; if either v_1 or v_k has a neighbor outside of $\{v_2, \dots v_{k-1}\}$, we can directly extend the path.

(ii) Assume that all neighbors of v_1 and v_k are in $\{v_2, \dots v_{k-1}\}$. Show that there is a pair v_i, v_{i+1}, s.t. $\{v_1, v_{i+1}\} \in E$ and $\{v_i, v_k\} \in E$. *(Hint: pigeonhole principle)*

(iii) Show that if the above happens, there is actually a cycle on vertices $\{v_1, \dots v_k\}$. Use this to show that we can find a path of length $k + 1$ in G.

(iv) Conclude that every graph with minimum degree $\delta(G) \geq |V|/2$ has a Hamiltonian path.

(v) Use the same argument as in (ii) to deduce that G in fact has a Hamiltonian cycle.